

MAT1001 Final Examination

Monday, December 18, 2023

Time: 1:30 - 4:30 PM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The maximum score of this examination is **162**.*
3. *There are **14** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Answer all questions on the **answer book**.*
6. *Show your intermediate steps **except Questions 1 and 2** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *The symbols $\arcsin$, $\arctan$, etc., denote the inverse trigonometric functions $\sin^{-1}$, $\tan^{-1}$, etc. .*
8. *Express irrational numbers in exact forms instead of decimal forms; e.g., write $\sqrt{2}$ instead of 1.414..., and write $\ln 2$ instead of 0.693....*

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MAT1001 Final Exam Questions

1. [12] True (T) or False (F)? No justification is required.

- (i) The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x^{4/3}}$ does not exist.
- (ii) Supposing the functions $f(x)$ and $g(x)$ are positive for sufficiently large x , if f and g grow at the same rate as $x \rightarrow \infty$, then $f = O(g)$ and $g = O(f)$ as $x \rightarrow \infty$.
- (iii) $\frac{1}{x} - \frac{1}{x^2} = o\left(\frac{1}{x}\right)$ as $x \rightarrow \infty$.
- (iv) $\int_{-\infty}^{\infty} x \, dx = 0$.
- (v) Assume that we use n evenly spaced subintervals of $[a, b]$ to approximate a definite integral over $[a, b]$. Then Simpson's rule with n subintervals always provides a smaller approximation error comparing to the trapezoidal rule with n subintervals.
- (vi) Every autonomous differential equation has at least one equilibrium value.

2. [18] Short questions: no intermediate step is required.

- (i) If $y = \int_{\tan x}^0 \frac{1}{1+t^2} dt$, then dy/dx is ().
- (ii) Let $f(x) = x^3 - 3x^2 - 1$, $x \geq 2$. Then the value of $\frac{df^{-1}}{dy}$ at the point $y = 15 = f(4)$ is ().
- (iii) Let y be a function of x that satisfies $yy' = -x$ and $y(0) = -1$. Which of the following could be the graph of the function?
 - A. A straight line.
 - B. A circle.
 - C. The upper half of a semicircle.
 - D. The lower half of a semicircle.
- (iv) The equation $2xy + y^2 = x + y$ defines a curve on the xy -plane. The slope of the tangent line to the curve at $(x, y) = (0, 0)$ is ().

(v) The linearization of $f(x) = \sin(2x)$ centred at $x = 0$ is $L(x) = (\quad)$.

(vi) Find the partial fraction decomposition for $\frac{2}{x^3 - x^2 + x - 1}$.

3. [15] Evaluate the following limits.

(i) $\lim_{x \rightarrow 0} \frac{\sqrt[2]{4+x} - \sqrt[3]{8+x}}{x}$.

(ii) $\lim_{x \rightarrow \infty} \frac{e^{2x} \arctan(e^x)}{e^{2x} + x}$.

(iii) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \ln \left(\sqrt[n]{1 + \frac{k}{n}} \right)$.

4. [35] Evaluate the following integrals.

(i) $\int (x+10)(x-10)^{1/3} dx$.

(ii) $\int \frac{1}{9x^2 + 6x + 5} dx$.

(iii) $\int \frac{\ln x}{x^2} dx$.

(iv) $\int \cos(\ln x) dx$.

(v) $\int \tan^5 \theta \sec^7 \theta d\theta$.

(vi) $\int_2^3 \frac{1}{x^2 \sqrt{x^2 - 1}} dx$.

(vii) $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$.

5. [6] Find the length of the curve given by

$$x = (y/4)^2 - 2 \ln(y/4), \quad 4 \leq y \leq 12.$$

6. [6] Find the volume of the solid generated by revolving the curve

$$y = \frac{1}{\sqrt{1+x^2}}, \quad -\frac{\sqrt{3}}{3} \leq x \leq \sqrt{3}$$

about the x -axis.

7. [6] Suppose that x and y are related by the equation

$$y = \int_0^x \frac{1}{\sqrt{1+t^2}} dt.$$

Show that $\frac{d^2x}{dy^2} = Kx$ for some real number K , and find the value of K .

8. [10] Find the nonzero real constant a for which

$$\lim_{x \rightarrow 0} (\cos x)^{2a/(x^2)} = 2 \int_a^\infty x e^{-x} dx.$$

9. [7] Solve the differential equation with the initial condition:

$$\frac{dy}{dx} + xy^2 - 2y^2 = 2 - x, \quad y(0) = 1.$$

10. [7] Find the general solution to the following differential equation:

$$y' - \frac{\sin x}{1 + \cos x} y = \frac{x^2}{1 + \cos x}, \quad 1 + \cos x \neq 0.$$

11. [4+4+6] A particle moves along the x -axis with a velocity (in m/sec) of

$$v(t) = 2t - t^2, \quad t \geq 0.$$

It starts at the point $x = 0$ at time $t = 0$ (sec), and the velocity is positive in the positive x -direction.

- (i) Compute the distance it traveled over the interval $0 \leq t \leq 3$ sec.
- (ii) Compute its x -coordinate at time $t = 3$ sec.
- (iii) Supposing a resistive force of $F(t)$ (in N) acts on the particle where

$$F(t) = -a v(t), \quad t \geq 0.$$

Here, $a > 0$ is a constant and the negative sign indicates that the force $F(t)$ acts in a direction opposite to the velocity $v(t)$. Compute the work done by the particle to overcome the force over the interval $0 \leq t \leq 3$ sec.

12. [6+4] The portion of the curve given by

$$y = \arctan(x), \quad 0 \leq x \leq a, \quad a > 0$$

is rotated around the y -axis to form a bowl facing up. All the lengths are measured in meters. Water is let in from the top at the constant rate of r (in m^3/sec) beginning at time $t = 0$ sec at which time point the bowl is empty. Assume that $\arctan(a) > \pi/4$.

- (i) Find the time T sec it takes for the water level to reach the height of $y = \pi/4$ meters.
- (ii) At time $t = T$ sec, find the rate at which the water level is rising.

13. [5+5] For each of the following improper integrals, either find its value, or show that it diverges.

- (i) $\int_0^\infty \frac{3^x}{1 + 3^{2x}} dx.$
- (ii) $\int_0^1 \frac{\ln^2(x+1) \sqrt[3]{x^2 + x \sin x}}{x^4} dx.$

14. [6] Supposing

$$\int_0^4 e^{(x-2)^4} dx = K,$$

find the value of

$$\int_0^4 x e^{(x-2)^4} dx.$$

In the notation above, $(x-2)^4$ is the exponent of e .